

Error detection & Correction

Date :

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Hamming Code:

- 1-bit errors can be identified & corrected
- developed by R. W. Hamming

Data (m) $\rightarrow$ any length

→ redundant bits (r) $\rightarrow$ length of r

$$2^r \geq m + r + 1$$

Senderside $\rightarrow m + r$

All bit positions placed at powers of 2 $\Rightarrow (2^1, 2^2, 2^3, \dots)$ marked as parity

$r_1 \Rightarrow 2^0$ (1st pos)

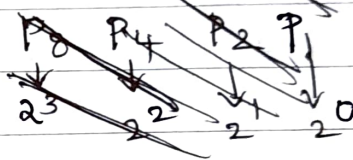
$r_2 \Rightarrow 2^1$ (2nd pos)

$r_3 \Rightarrow 2^2$ (4th pos)

$r_4 \Rightarrow 2^3$ (8th pos)

remaining bits $\Rightarrow$ data

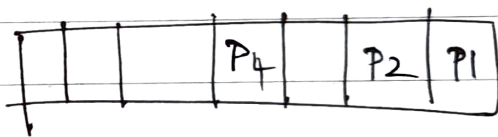
ex - $r = 4$



ex:

① Data (M) = 1001

(r) = ?



$$2^r \geq m + r + 1$$

$$2^1 \geq 4 + 1 + 1$$

$$2 \geq 6 \quad \times$$

$$2^2 \geq 4 + 2 + 1$$

$$4 \geq 7 \quad \times$$

$$2^3 \geq 4 + 3 + 1$$

$$8 \geq 8 \quad \checkmark$$

$M = 1001$

$r = 3$ bits

$r = 3 \text{ bits}$

$P_4 \ P_2 \ P_1$

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7	6	5	4	3	2	1
1	0	0	P_4	1	P_2	P_1

$P_1, P_2, P_4 = ?$
 $P_4 P_2 P_1$

1	001
2	010
3	011
4	100
5	101
6	110
7	111

P_1 position is '1'

P_1 is identified by first position is '1'

$P_1 \rightarrow 1, 3, 5, 7 \rightarrow \textcircled{P_1} 101 \rightarrow$

$P_2 \rightarrow 2, 3, 6, 7 \rightarrow \textcircled{P_2} 101 \rightarrow$

$P_4 \rightarrow 4, 5, 6, 7 \rightarrow \textcircled{P_4} 001 \rightarrow$

Considering
even parity

1's even $\rightarrow 0$

1's odd $\rightarrow 1$

$P_1 \Rightarrow P_1 101 \Rightarrow 0$

$P_2 \Rightarrow P_2 101 \Rightarrow 0$

$P_4 \Rightarrow P_4 001 \Rightarrow 1$

$P_1 = 0 ; P_2 = 0 ; P_4 = 1$
redundant bits

Sender side

{	7	6	5	4	3	2	1
	1	0	0	1	1	0	0

Receiver side

	6	5	4	3	2	1
1	0	0	1	1	0	0

during the transmission the data sent is

7 6 5 4 3 2 1

1 1 0 1 1 0 0 $\Rightarrow$ data with error

$P_1 \Rightarrow 1, 3, 5, 7 \Rightarrow 0101 \Rightarrow P_1$ should be zero

$P_2 \Rightarrow 2, 3, 6, 7 \Rightarrow 0111 \Rightarrow P_2$ should be 1

$P_4 \Rightarrow 4, 5, 6, 7 \Rightarrow 1011 \Rightarrow P_4$ should be 1

$$P_1 = 0$$

$$P_2 = 1$$

$$P_4 = 1$$

$$\left\{ \begin{array}{l} 0101 = 0 \\ 0111 = 1 \\ 1011 = 1 \end{array} \right.$$

$$P_4 P_2 P_1 = (110)_2 = (6)_{10} \text{ decimal}$$

In 6th position error

Receiver side

7	6	5	4	3	2	1
1	0	0	1	1	0	0

$P_1 = 1, 3, 5, 7 \Rightarrow 0101 \Rightarrow P_1 = 0$

$P_2 = 2, 3, 6, 7 \Rightarrow 0101 \Rightarrow P_2 = 0$

$P_4 = 4, 5, 6, 7 \Rightarrow 1001 \Rightarrow P_4 = 0$

$$P_1 = 0$$

$$P_2 = 0$$

$$P_4 = 0$$

no error

data = 1011

$r = 3$

7 6 5 4 3 2 1
1 0 1 P₄ P₂ P₁

$$P_1 = 1, 3, 5, 7 = P_1 111 \Rightarrow P_1 = 1$$

$$P_2 = 2, 3, 6, 7 = P_2 101 \Rightarrow P_2 = 0$$

$$P_4 = 4, 5, 6, 7 = P_4 101 \Rightarrow P_4 = 0$$

Sender
side

1 0 1 0 1 0 1

Receiver side:

7 6 5 4 3 2 1
1 0 1 0 1 0 1

$$P_1 = 1, 3, 5, 7 = 1111 \checkmark$$

$$P_2 = 2, 3, 6, 7 = 0101 \checkmark$$

$$P_4 = 4, 5, 6, 7 = 0101 \checkmark$$

Received
Correct data

Receiver side
with error

7 6 5 4 3 2 1
1 1 1 0 1 0 1

$$P_1 = 1, 3, 5, 7 = 1111$$

$$P_2 = 2, 3, 6, 7 = \underline{0}111$$

$$P_4 = 4, 5, 6, 7 = 0\underline{1}11$$

error

$$P_1 = 1111 - 0$$

$$P_2 = 0111 - 1$$

$$P_4 = 0111 - 1$$

$$\cancel{P_4} P_4 P_2 P_1$$

$$= (110)_2$$

$$= (6)_{10}$$

error in 6th
position

Flow Control

Flow Control is the process of managing the rate of data transmission b/w 2 nodes to prevent a faster sender from overwhelming a slow receiver

Protocols

for noiseless channel

- 1) simplest
- 2) stop & wait

for noisy channel

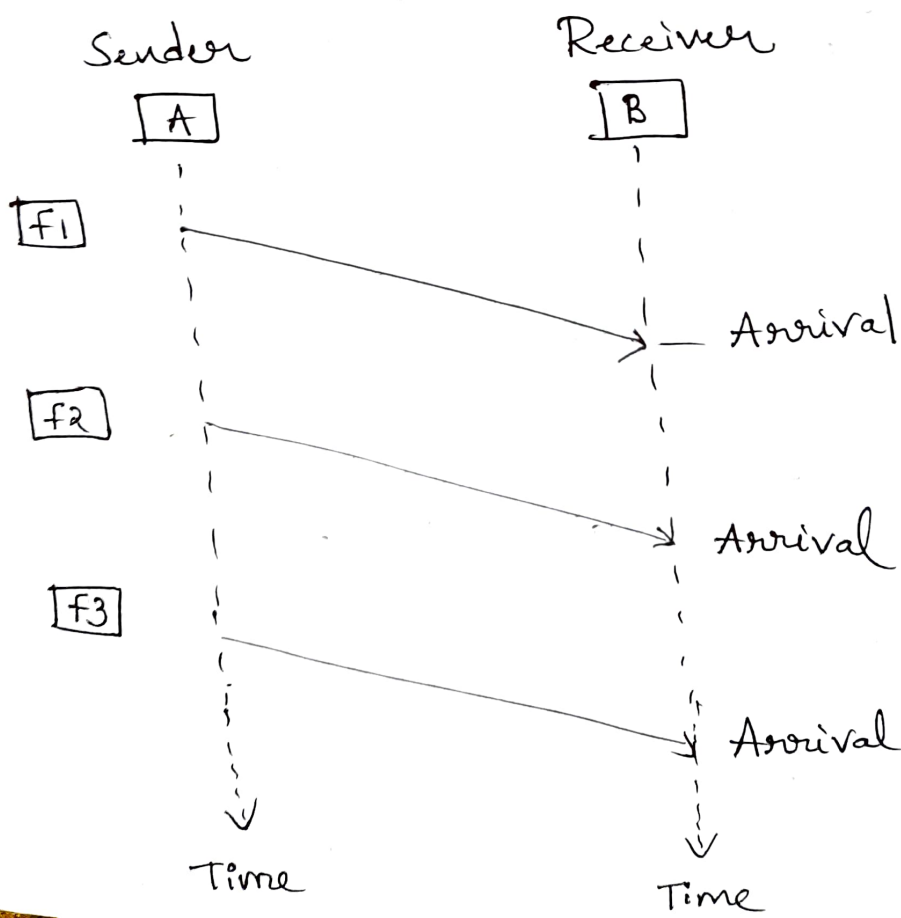
- 1) stop & wait - ARQ
- 2) Go-back-N ARQ
- 3) selective Repeat ARQ

① simplest protocol

(no flow or error control)

→ It allows the Unidirectional transmission of data frames

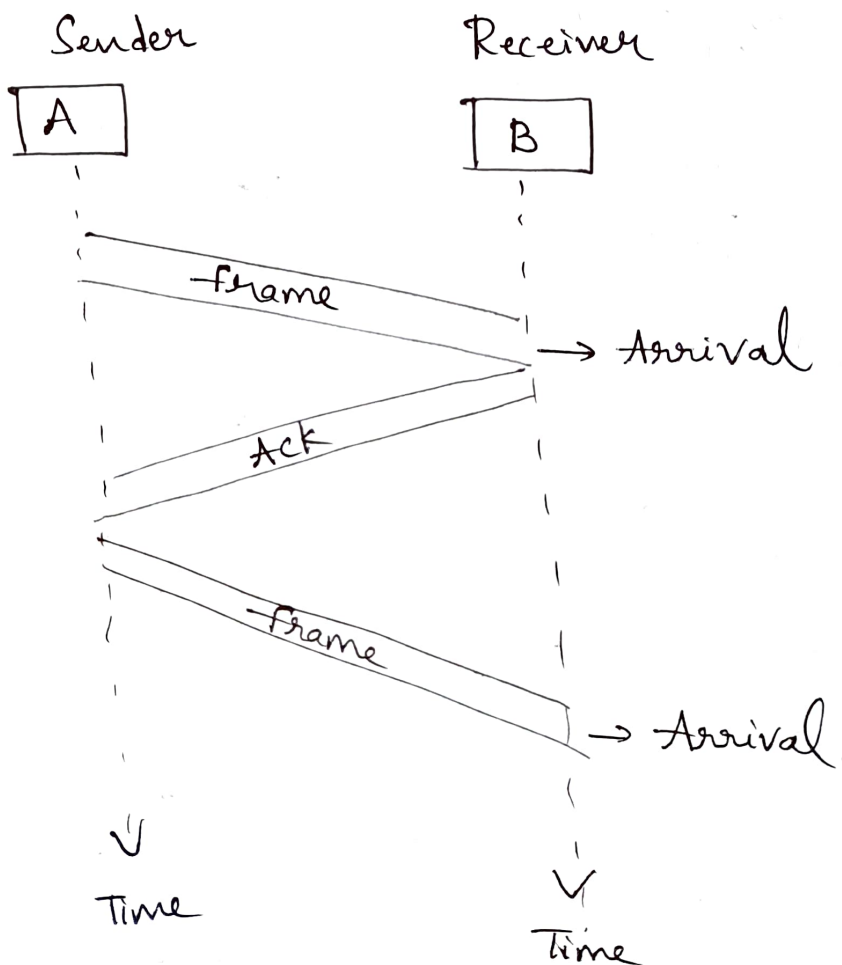
→ ~~the~~ receiver can immediately handle any frame it receives.



In this protocol none of the frames are lost during transmission.

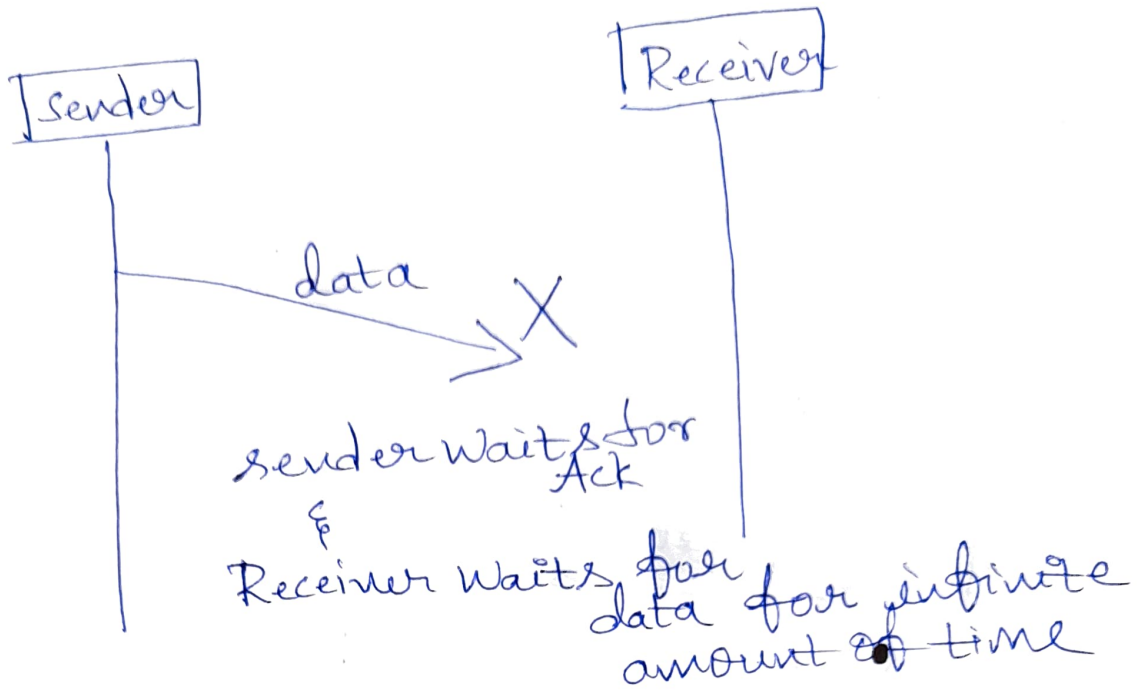
Stop & Wait Protocol:

- Generally, the receiver doesn't have enough storage space, if so the receiver might discard the received frames.
- This protocol makes sure, When a frame is sent by a sender, the sender stops until it receives Confirmation from the receiver.
- this protocol also follows Unidirectional data transmission but the receiver provides Confirmation by sending "Ack" frame (acknowledgement)
- therefore, a half duplex link is required

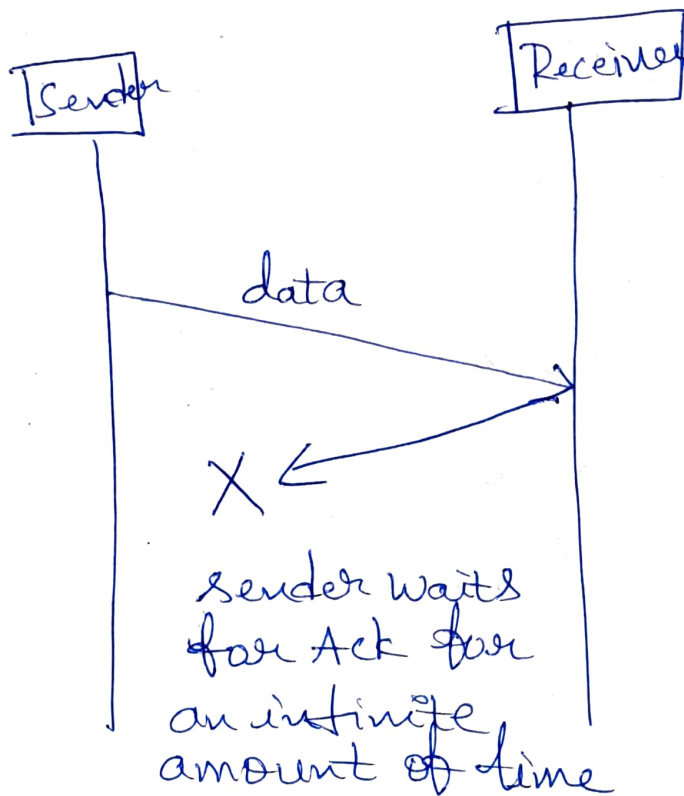


problems in stop & wait protocol

① Data lost



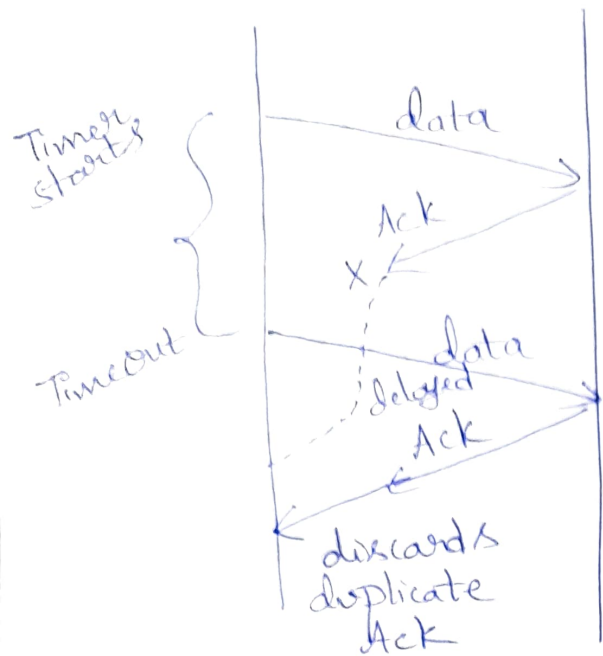
② Ack lost



③ delayed ack/data

Sender

Receiver

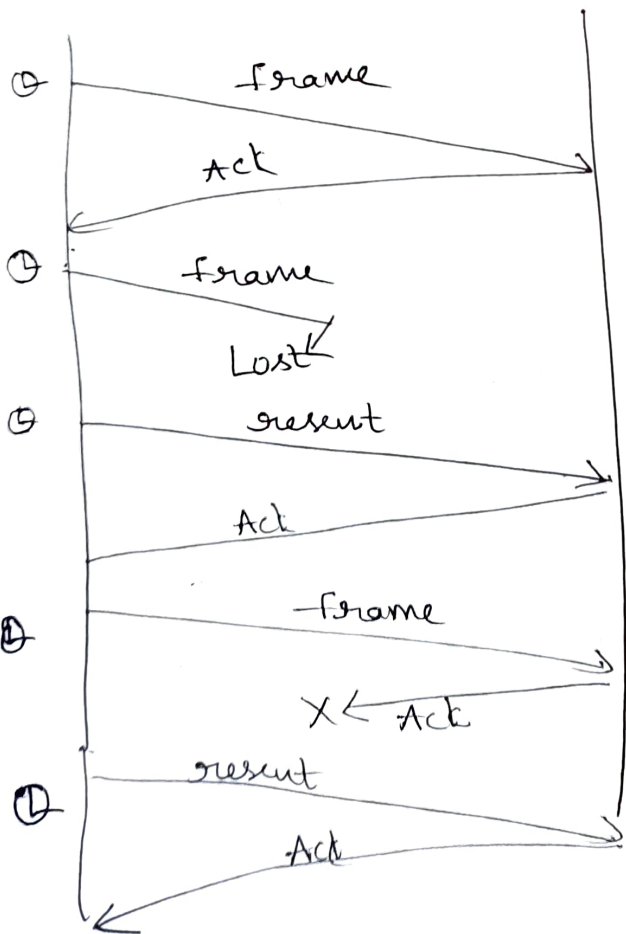


- 1) timer(Send-Ack)
- 2) sender needs a copy of the frame till its ack arrives

Duplicate packets

Sender

Receiver



- 1st frame
- Sent - Ack
- 2nd
- timeout resent
- 3rd
- Sent -> ack (lost)
- resent - ack (lost)

Stop & Wait protocol gives us an idea to add flow control to its predecessor, where channels are nonexistent.

We need to add flow control to our protocols

- └ Stop & wait ARQ
- └ Go-Back-N ARQ
- └ Selective repeat

① Stop & Wait ARQ :

ARQ → Automatic Repeat Request
→ stop & wait ARQ adds a simple error control mechanism to the stop & wait protocol.

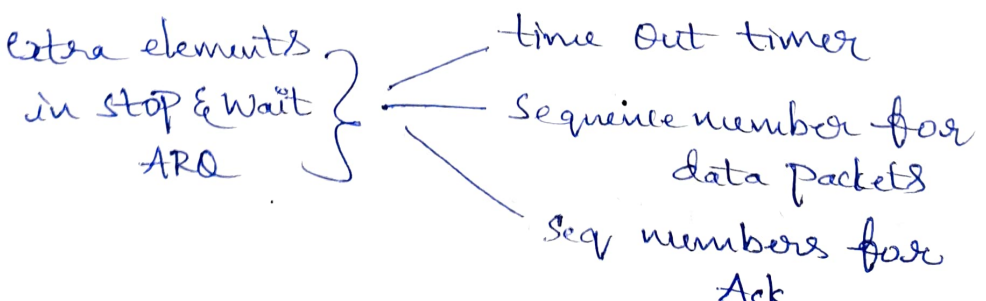
⇒ lost frames are more difficult to handle than corrupted frames in stop & wait protocol

1) no way to identify a frame (Correct one, duplicate or a frame out of order)

Solution : number the frames

Ack frames can also be corrupted and lost, so need sequence number for Ack also

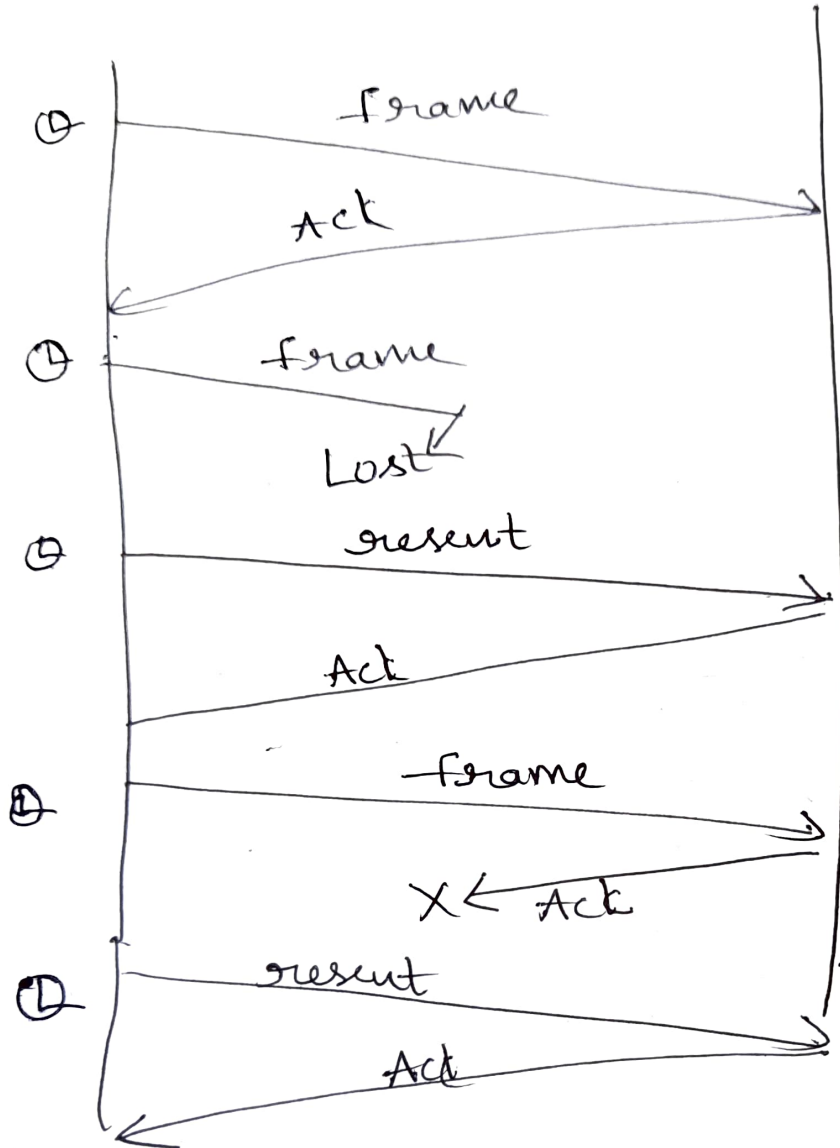
Error Correction in stop & wait ARQ is done by keeping a copy of the sent frame & retransmitting of the frame when timer expires.



Duplicate Packets

Sender

Receiver



1st
frame
Sent - Ack

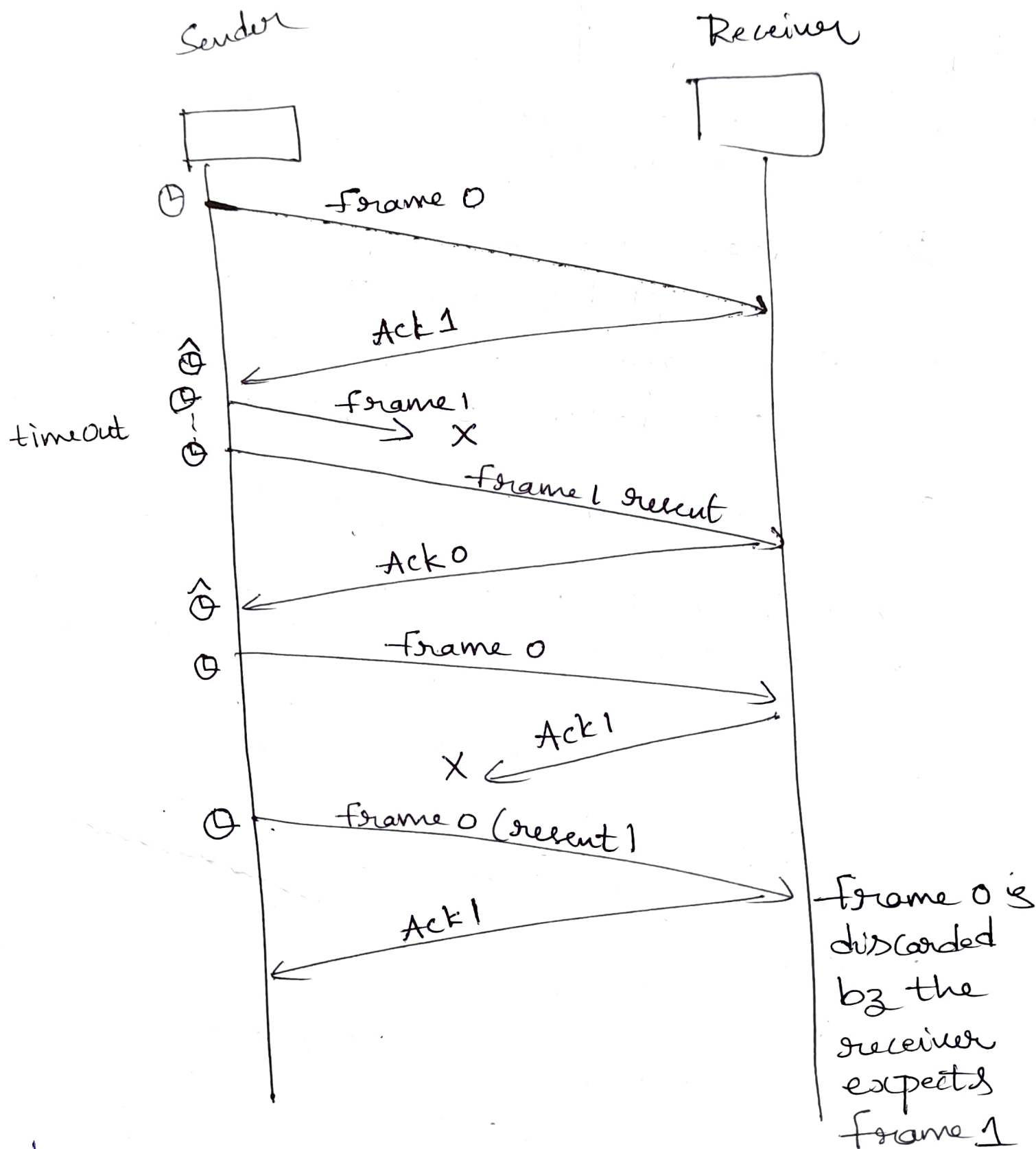
2nd
timeout resend

3rd
Sent → ack (lost)
resend - ack (frame)

duplicate
Ack

number of the next 0

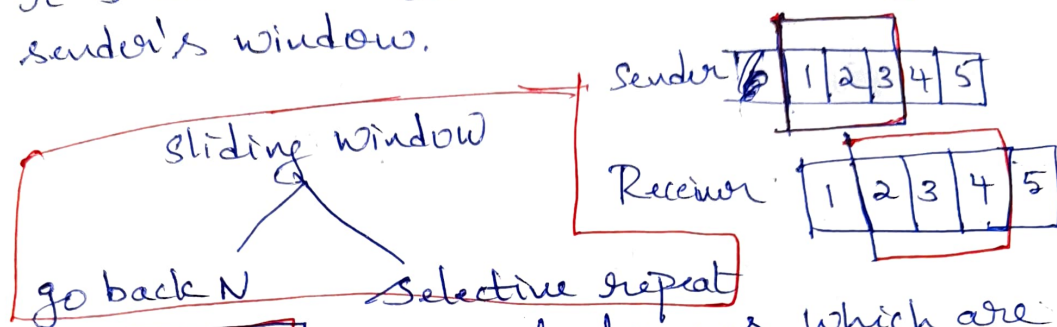
Sequence numbers 0, 1, 0, 1, 0, 1 ...
Ack numbers 1, 0, 1, 0, 1, 0 -



disadv: less efficiency

Sliding window protocol:

- uses concept of ~~pipelining~~ pipelining
- flow control protocol
- it allows the sender to send multiple frames before needing the ack's
- sender slides its window on receiving the ack's for the sent frames.
- This allows the sender to send more frames.
- It is called so bcz it involves sliding of sender's window.



Sending window: range of frames which are the concern of the sender

Go-back-N automatic repeat request:

- we can send frame (several) before receiving ack's
 - we keep a copy of these frames until the ack's arrive.
 - if any frame is lost, all subsequent frames have to be sent again.
- Sequence Numbers:

- ⇒ frames from a sending station are numbered sequentially.
- ⇒ include seq number for each frame in the header.

Sequence numbers ranging from

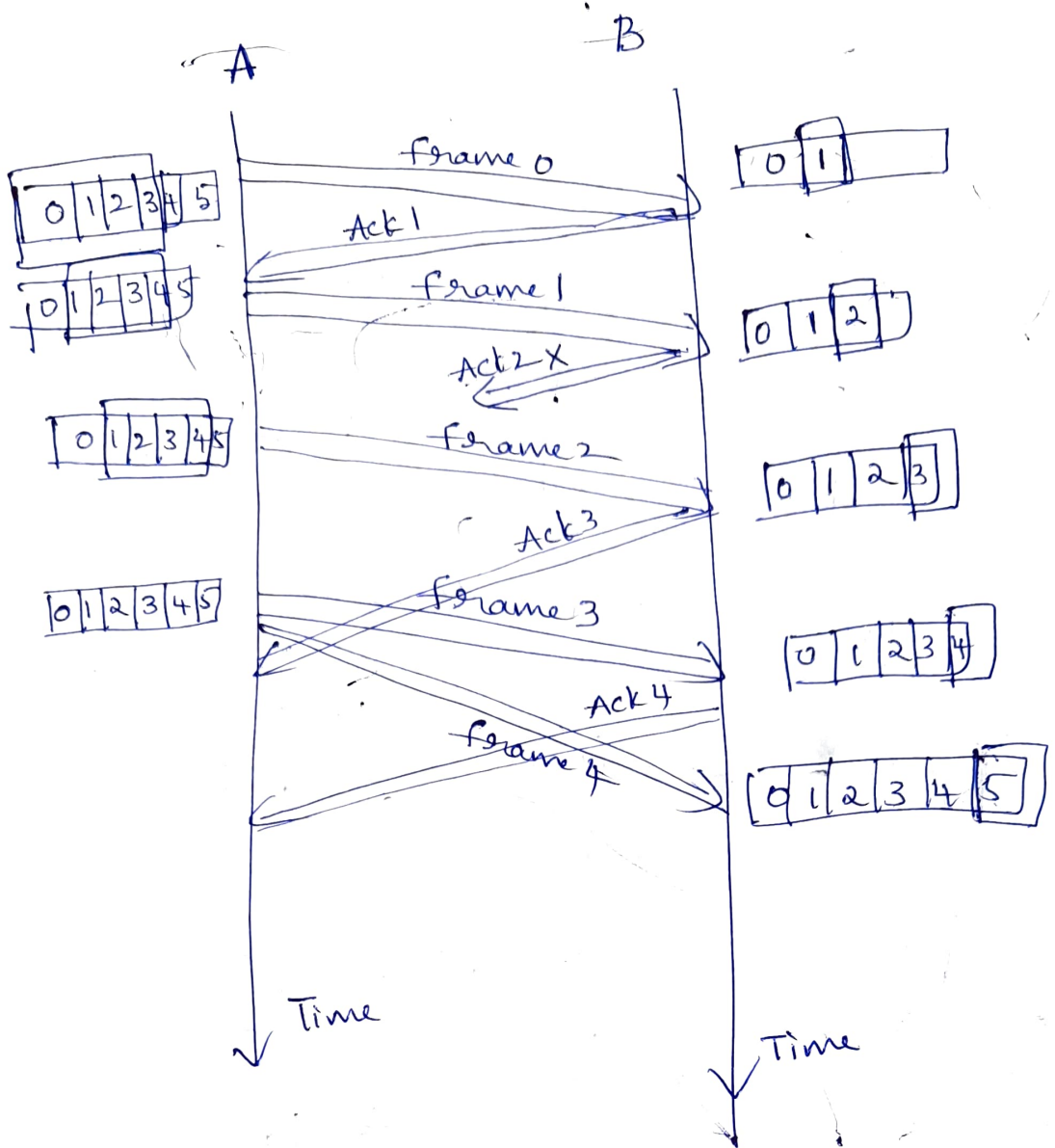
$$2^m - 1$$

$$m = 4 \quad 2^4 - 1 = 15$$

0, 1, 2, 3, 4, ..., 15

no header allow m bits for seq no
 m = size of the seq no. field.

Cumulative Ack example

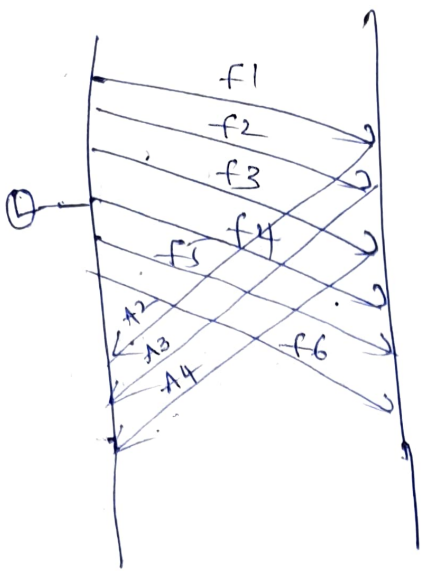


→ In above example forward channel is reliable, but reverse is not.

⇒ No dataframes are lost, but some Ack's are delayed & one is lost

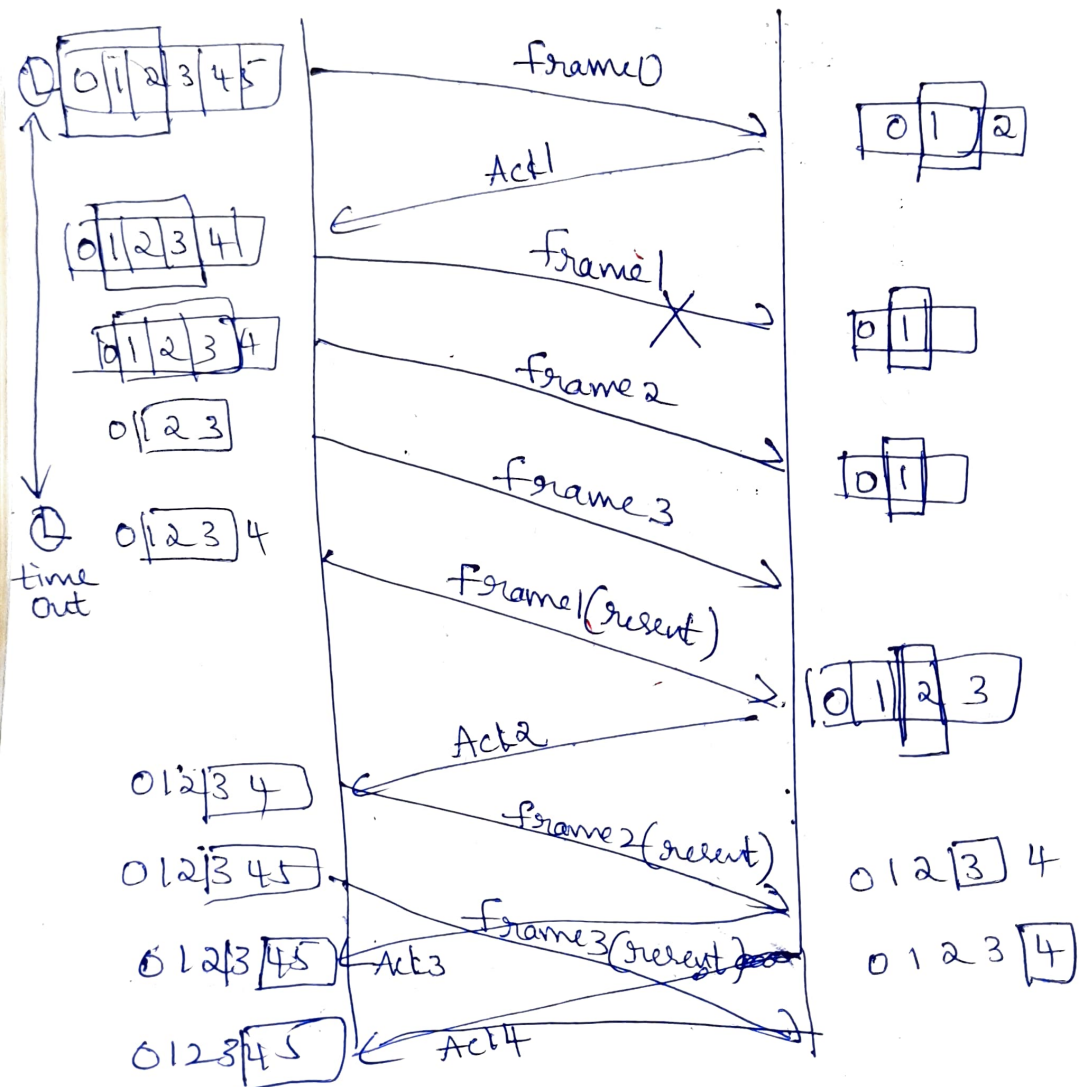
⇒ above example shows how cumulative ack can help if ack's are delayed or lost.

Ack 2 is lost but Ack 3 serves as both Ack 2 & Ack 3



if timer expires at f4
need to resend
4, 5, 6 ~~again~~
frames again

frames discarded b3 out of order

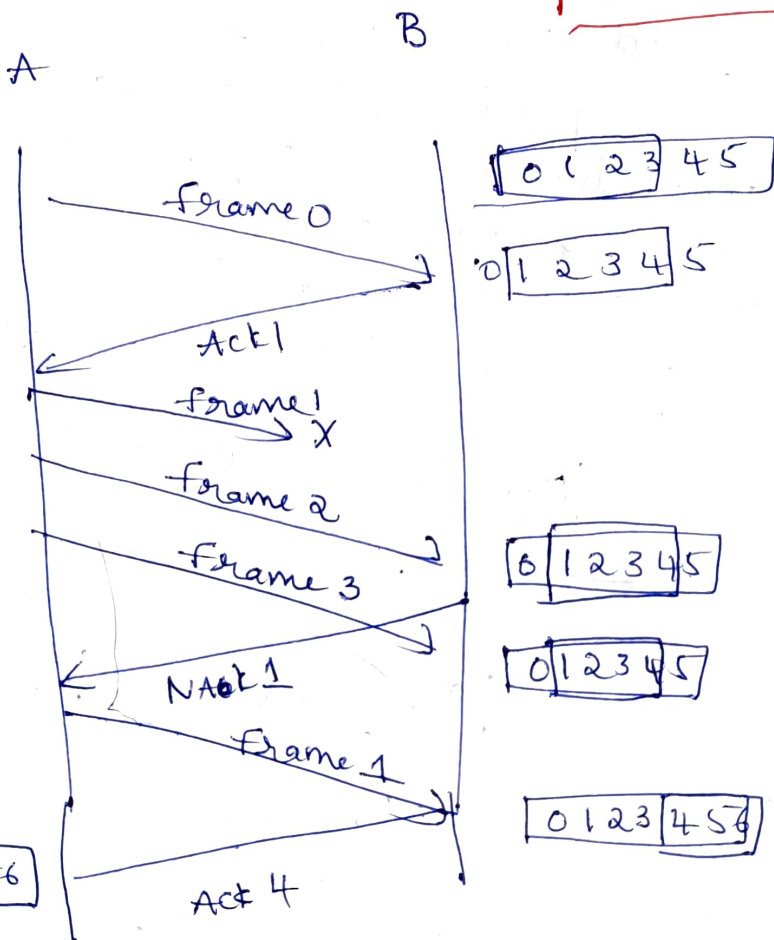


- frames 0, 1, 2, 3 sent
- frame '1' is lost
- receiver receives 2, 3 discarded b/c out of order

Selective Repeat ARQ

- doesn't resend all frames
- doesn't resend N frames when just one frame is damaged; only the damaged frame is resent.
- Unlike Go-back N ARQ, Receiver window size is not always 1.

NAK → avoid unnecessary retransmissions



Piggybacking

→ technique to improve the efficiency of the bidirectional protocols.

When a frame is carrying data from A to B, it can also carry control information about arrived (or lost) frames from B
both

Media Access Control Sublayer

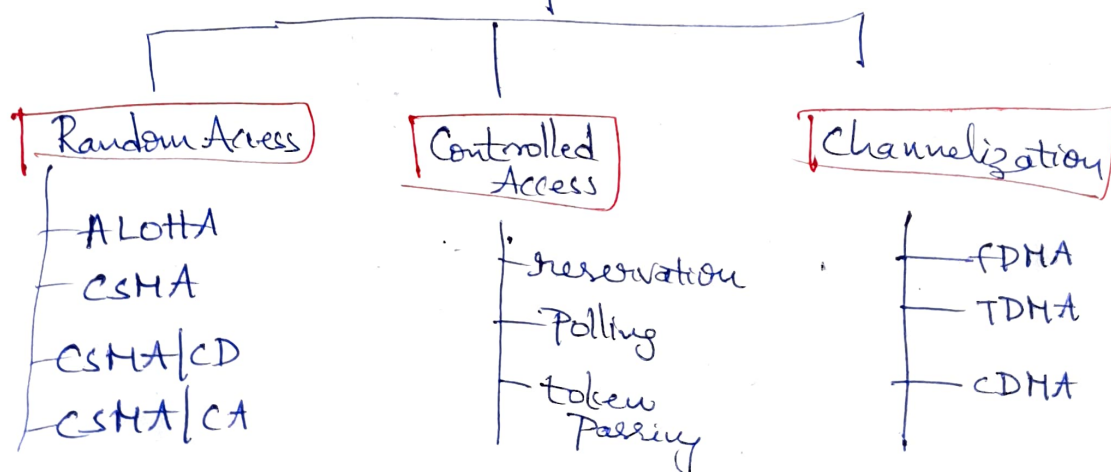
Data link layer has 2 sub layers

- 1) logical link control (LLC)
- 2) Media Access Control (MAC)

Functions of MAC sublayer:

- 1) It is responsible for encapsulating frames
- 2) It resolves addressing of source station & the destination station.
- 3) It resolves channel access issues
- 4) It performs collision resolution

Multiple Access protocols



Random Access Protocols:

Random Access or Contention methods.

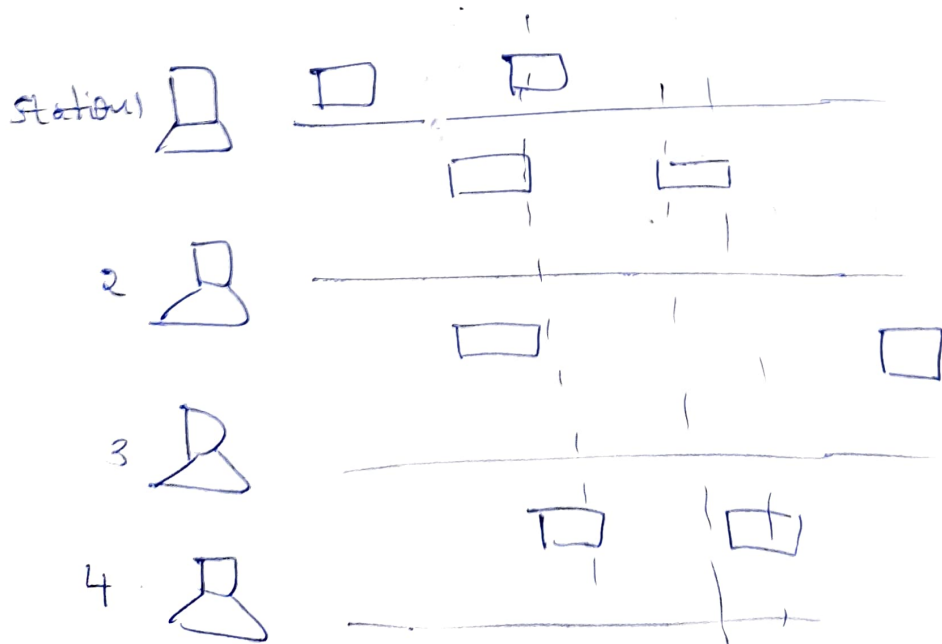
- No station is superior to another station & none is assigned control over another.
- there is no scheduled time for a station to transmit.
- transmission is random among stations.

ALOHA:

- developed by University of Hawaii
- designed for wireless LAN
- used on shared medium

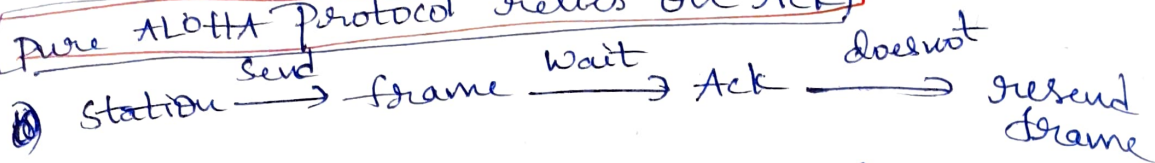
Pure ALOHA:

- original ALOHA is called pure ALOHA
- each station sends a frame whenever it has a frame to send. But there is only one channel to share $\Rightarrow$ possibility of collisions b/w frames from different station



only
2 frame
survive

Pure ALOHA Protocol relies on Ack

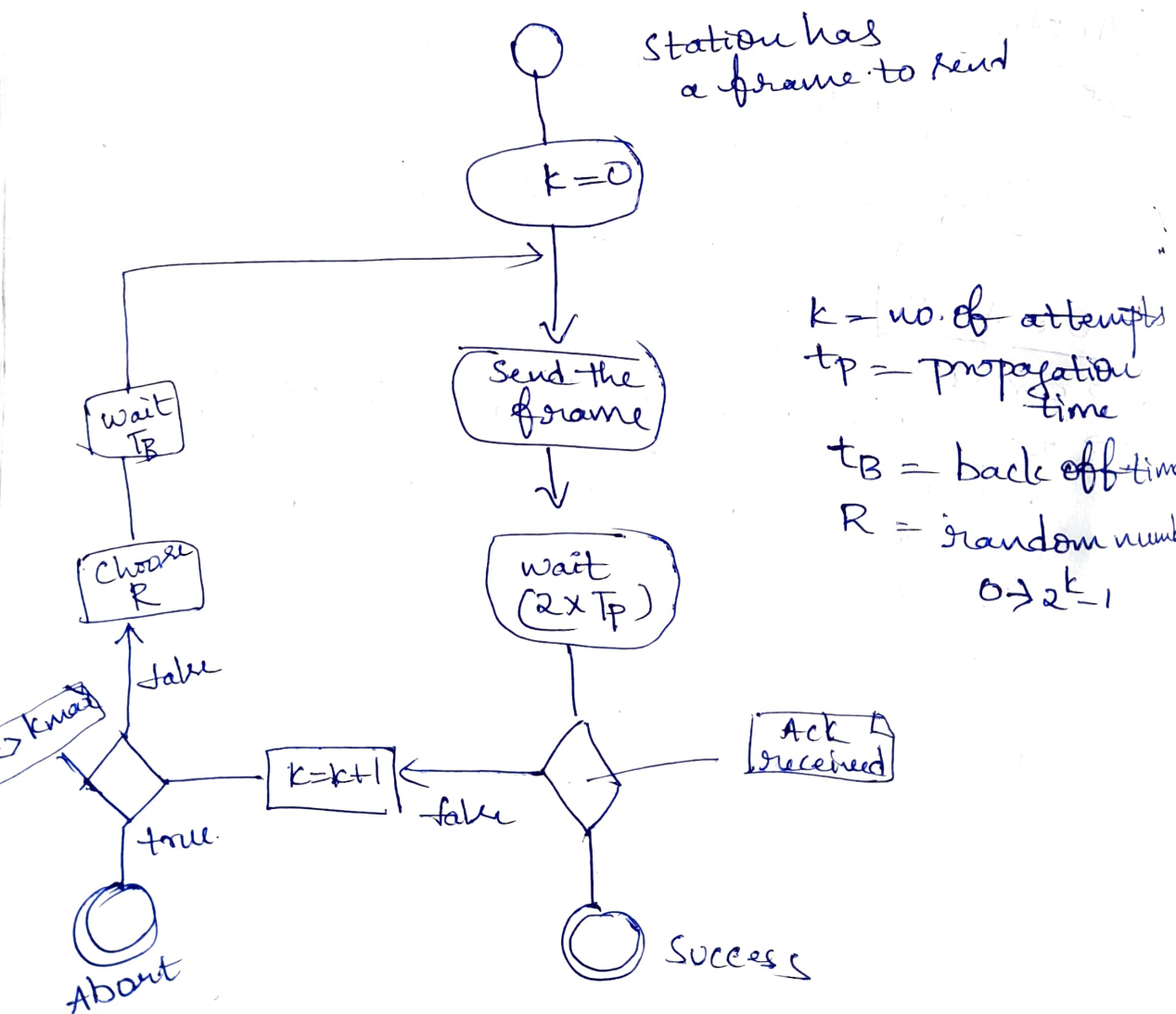


if all stations try to resend the data (after time out) will collide again

Sol: ① each station ~~wait~~ should wait a random amount of time before resending $\Rightarrow$ avoid collision

② after max no. of retransmission, a station must give up & try later

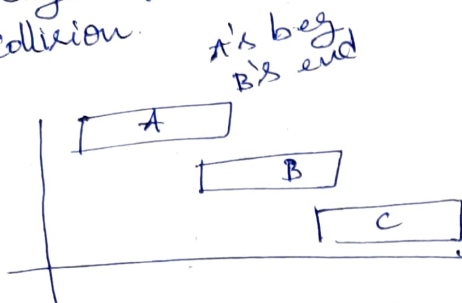
Procedure for pure ALOHA



T_{for} = (avg-transmission time)
average time required to send out a frame
 $k_{max} = 15$

Vulnerable time

length of time in which there is a possibility of collision.



$2 \times T_{for}$ Vulnerable time

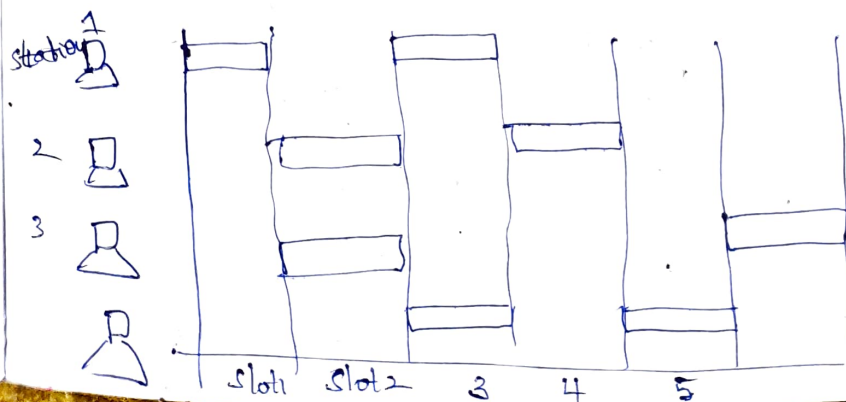
Throughput:

$$S = G \times e^{-2G}$$

G = avg no. of frames generated by the system during one frame transmission time

Slotted ALOHA

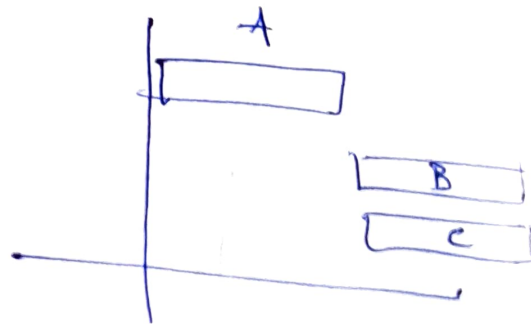
⇒ divide the time in to slots & force the stations to send only at the beginning of the time slot.



Slotted ALOHA

- ⇒ a station is allowed to send only at the beginning of the synchronized time slot, if a station misses this moment, it must wait until the beginning of the next slot.
- ⇒ But still the possibility of collision if two stations try to send at the beginning of the same time slot.
- ⇒ vulnerable time is now reduced to one-half equal to T_{fr}

$$\text{Throughput } - (S) = G \times e^{-2G}$$
$$\text{vulnerable time} = T_{fr}$$



Q) A pure ALOHA n/w transmits 200-bit frame on a shared channel of 200 kbps. What is the throughput if the system produces

- a) 1000 frames per second?
- b) 500 frames per second?
- c) 250 frames per second?

Sol:- Throughput for pure ALOHA is

$$S = G \times e^{-2G}$$

G = avg no. of frames generated by the system during one frame transmission time.

→ Compute one frame transmission time

$$T_{fr} = \frac{200 \text{ bits}}{200 \text{ kbps}} = \frac{200 \text{ bits}}{200 \times 10^3 \text{ bps}}$$

$$T_{fr} = 10^{-3} \text{ seconds}$$

$$T_{fr} = 1 \text{ milliseconds}$$

① 1000 frames per second

$$G = 1000 \text{ frames per second} \times T_{fr}$$
$$= 1000 \times 10^{-3}$$

$$G = 1$$

$$S = G \times e^{-2G} = 1 \times 0.135$$

$$S = 0.135$$

throughput = $1000 \times 0.135 = 135$ frames
135 frames per second. Successfully out of 1000 frames

$$2) T_{fr} = 1 \text{ ms}$$

$$G = 500 \text{ frames per second} \times 1 \text{ ms}$$

$$= 500 \times 10^{-3}$$

$$G = \frac{1}{2}$$

$$S = G \times e^{-2G}$$

$$= \frac{1}{2} \times e^{-2 \times \frac{1}{2}}$$

$$= \frac{1}{2} \times e^{-1}$$

$$= 0.5 \times 0.367$$

$$= 0.183$$

The system produces 500 frames

$$0.183 \times 500 = 91.96$$

ie 92 frames reach successfully out of 500 frames

$$3) T_{fr} = 1 \text{ ms}$$

$$G = 250 \times 1 \text{ ms}$$

$$= 250 \times 10^{-3} = \frac{1}{4}$$

$$S = G \times e^{-2G}$$

$$= \frac{1}{4} \times e^{-2 \times \frac{1}{4}}$$

$$= \frac{1}{4} \times e^{-1/2}$$

$$= 0.25 \times 0.606$$

$$= 0.151$$

The system produces 250 frames

$$0.151 \times 250 = 37.90$$

i.e. 38 frames reach successfully out of 250 frames.

Q) A Slotted ALOHA n/w transmits 200-bit frames using a shared channel with a 200-kbps bandwidth. Find the throughput if the system produces

- a) 1000 frames per second
- b) 500 frames per second
- c) 250 frames per second

Sol:- Throughput for slotted ALOHA

$$S = G \times e^{-G}$$

for G , compute T_{fr}

$$T_{fr} = \frac{200 \text{ bits}}{200 \text{ kbps}} = \frac{200 \text{ b}}{200 \times 10^3 \text{ bps}}$$

$$T_{fr} = 10^{-3} \text{ seconds}$$

$$T_{fr} = 1 \text{ ms}$$

$$\begin{aligned} a) G &= 1000 \text{ frames per second} \times T_{fr} \\ &= 1000 \times 1 \text{ ms} \\ &= 10^3 \times 10^{-3} \\ &= 1 \end{aligned}$$

$$\begin{aligned} S &= G \times e^{-G} = 1 \times 0.367 \\ &= 0.367 \end{aligned}$$

the system produces 1000 frames per second
 $= 0.367 \times 1000$
 $= 367$ frames out of 1000 frames can
reach successfully.

$$b) G = 500 \times T_{fr}$$

$$= 500 \times 10^{-3}$$

$$G = \frac{1}{2}$$

$$S = G \times e^{-G} = \frac{1}{2} \times e^{-1/2}$$

$$= 0.5 \times 0.606$$

$$= 0.303$$

The system produces 500 frames per
second.

$$500 \times 0.303 = 151.6$$

152 frames out of 500 frames ~~per second~~
can reach successfully

$$c) G = 250 \times T_{fr} = 250 \times 10^{-3} = \frac{1}{4}$$

$$S = G \times e^{-G} = \frac{1}{4} \times e^{-1/4}$$

$$= 0.25 \times 0.77$$

$$= 0.195$$

The sys produces 250 frames per second
 $250 \times 0.195 = 48.6$

49 frames out of 250 frames can
reach successfully.